

$$\int_a^b f(x)g'(x) dx = f(x)g(x) \Big|_a^b - \int_a^b f'(x)g(x) dx$$

$$\begin{aligned} \int_0^{\pi/2} x \sin x dx &= \int u dv = uv - \int v du \\ &= x \cos x - \int \cos x dx \\ &= x \cos x + \sin x + C \end{aligned}$$

8.5 Integration by partial fractions

Goal: $\int \frac{p(x)}{q(x)} dx$, $p(x), q(x)$ polynomials

$$\begin{aligned} \text{E.g. } \int \frac{92x}{x^2-x-2} dx &= \int \frac{92x}{(x-2)(x+1)} dx = \int \left(\frac{A}{x-2} + \frac{B}{x+1} \right) dx \\ &= A \int \frac{1}{x-2} dx + B \int \frac{1}{x+1} dx \\ &= A \ln|x-2| + B \ln|x+1| + C \end{aligned}$$

Theorem: Consider ratio function $\frac{p(x)}{q(x)}$. Suppose degree of $p(x) <$ degree of $q(x)$ (If not, carry out long division)

(i) $q(x)$ can be factorized into $q(x) = c(x-r_1)^{m_1}(x-r_2)^{m_2} \dots (x-r_k)^{m_k}(x^2+b_1x+c_1)^{n_1} \dots (x^2+b_lx+c_l)^{n_l}$; $x^2+b_i+c_i$ irreducible $\forall i=1, \dots, l$

(ii) $\frac{p(x)}{q(x)}$ can be composed into

$$\frac{p(x)}{q(x)} = \frac{A_{1,1}}{x-r_1} + \frac{A_{1,2}}{(x-r_1)^2} + \dots + \frac{A_{1,m_1}}{(x-r_1)^{m_1}} + \dots + \frac{A_{k,1}}{x-r_k} + \dots + \frac{A_{k,m_k}}{(x-r_k)^{m_k}} + \frac{B_{1,1}x+C_{1,1}}{x^2+b_1x+c_1} + \dots + \frac{B_{l,1}x+C_{l,1}}{x^2+b_lx+c_l} + \dots + \frac{B_{l,n_l}x+C_{l,n_l}}{(x^2+b_lx+c_l)^{n_l}}$$

UGLY PRETTY

$$8.5/8 \quad \frac{t^4+9}{t^4+9t^2} = 1 - \frac{9t^2-9}{t^4+9t^2} = 1 - 9 \cdot \frac{(t^2-1)}{t^4+9t^2}$$

$$t^4+9t^2 = t^2(t^2+9) \rightarrow \frac{t^2-1}{t^4+9t^2} = \frac{A}{t} + \frac{B}{t^2} + \frac{Cx+D}{t^2+9}$$

$$t^2-1 = (t^2+9t)A + (t^2+9)B + t^2(Cx+D)$$

$$A+C=0, B+C+D=1, 9A=0, 9B=-1$$

$$A=0, B=-\frac{1}{9}, D=\frac{10}{9}$$

$$\therefore \frac{t^4+9}{t^4+9t^2} = 1 - 9 \left(-\frac{1}{9(t^2+9)} + \frac{10}{9t^2} \right)$$

$$= 1 + \frac{1}{t^2} - \frac{10}{t^2+9}$$

$$8.5/14 \quad \int_{1/2}^1 \frac{y+4}{y^2+y} dy = \int_{1/2}^1 \left(\frac{A}{y+1} + \frac{B}{y} \right) dy$$

$$Ay + B(y+1) = y+4 \rightarrow A+B=1, B=4 \rightarrow A=-3$$

$$\begin{aligned} \therefore \int_{1/2}^1 \frac{y+4}{y^2+y} dy &= \int_{1/2}^1 \left(\frac{-3}{y+1} + \frac{4}{y} \right) dy \\ &= [3 \ln(y+1)]_{1/2}^1 + [4 \ln y]_{1/2}^1 \\ &= -3 \ln 2 + 3 \ln \frac{3}{2} + 4 \ln 1 - 4 \ln \frac{1}{2} \\ &= 3 \ln \frac{3}{2} + 4 \ln 2 \end{aligned}$$

8.4 Trigonometric substitution

Recall: $\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$

$$\int \frac{1}{1+x^2} dx = \arctan x$$

$$\int \frac{1}{x\sqrt{x^2-1}} dx = \operatorname{arcsec} |x| + C$$

Integrals involving $\sqrt{a^2-x^2}$, $a > 0$ const

Idea: $|x| \leq a \rightarrow \left| \frac{x}{a} \right| \leq 1, -1 \leq \frac{x}{a} \leq 1$. Let $\frac{x}{a} = \sin \theta, \theta = \sin^{-1}\left(\frac{x}{a}\right)$ $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ $\cos \theta \geq 0$ PRETTY

$$\therefore \sqrt{a^2-x^2} = \sqrt{a^2-a^2\sin^2\theta} = a|\cos\theta| = a\cos\theta$$

UGLY

$$8.4/20 \quad \int \frac{\sqrt{9-w^2}}{w^2} dw, \quad w = 3\sin\theta \rightarrow dw = 3\cos\theta d\theta$$

$$\theta = \sin^{-1}\left(\frac{w}{3}\right)$$

$$\int \frac{\sqrt{9-w^2}}{w^2} dw = \int \frac{3\cos\theta}{9\sin^2\theta} \cdot 3\cos\theta d\theta$$

$$= \int \cot^2\theta d\theta = \int (\csc^2\theta - 1) d\theta$$

$$= -\cot\theta - \theta + C = -\frac{\sqrt{9-w^2}}{w} - \arcsin\left(\frac{w}{3}\right) + C$$

$\cot\theta = \frac{\cos\theta}{\sin\theta} = \frac{\sqrt{1-\frac{w^2}{9}}}{\frac{w}{3}} = \frac{\sqrt{9-w^2}}{w}$

Integrals involving $\sqrt{a^2+x^2}$; $-\infty < \frac{x}{a} < \infty$

$$1 + \tan^2\theta = \sec^2\theta$$

Let $X = a\tan\theta \rightarrow \sqrt{a^2+x^2} = \sqrt{a^2+a^2\tan^2\theta} = a|\sec\theta| = a\sec\theta$

$$-\frac{\pi}{2} < \theta < \frac{\pi}{2} \rightarrow \sec\theta > 0$$

$$\theta = \arctan\left(\frac{x}{a}\right), \quad dx = \sec^2\theta d\theta$$

$$8.4/18' \quad \int \frac{1}{x^2\sqrt{x^2+9}} dx \quad X = 3\tan\theta \rightarrow dx = 3\sec^2\theta d\theta$$

$$\theta = \arctan\left(\frac{x}{3}\right) \quad \sqrt{x^2+9} = 3\sec\theta \quad u = \sin\theta \rightarrow du = \cos\theta d\theta$$

$$\hookrightarrow \int \frac{1}{9\tan^2\theta \cdot 3\sec\theta} \cdot 3\sec^2\theta d\theta = \int \frac{\sec\theta}{9\tan^2\theta} d\theta = \int \frac{\cos\theta}{9\sin^2\theta} d\theta = \int \frac{1}{9u^2} du = -\frac{1}{9u} + C$$

$$\sin\left(\arctan\left(\frac{x}{3}\right)\right) = \frac{\sqrt{x^2+9}}{3x} = -\frac{\sqrt{x^2+9}}{9x} + C$$

Integrals involving $\sqrt{x^2 - a^2} \rightarrow |x| \geq a$

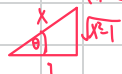
Let $X = a \sec \theta$: $\theta \in [0, \pi] - \{\frac{\pi}{2}\}$

$$\therefore \theta = \operatorname{arcsec}\left(\frac{X}{a}\right), dX = a \sec \theta \tan \theta d\theta$$

$$\sqrt{x^2 - a^2} = \sqrt{a^2 \sec^2 \theta - a^2} = a |\tan \theta| = \begin{cases} a \tan \theta & ; 0 \leq \theta < \frac{\pi}{2} \\ -a \tan \theta & ; \frac{\pi}{2} < \theta \leq \pi \end{cases}$$

8.4/25 $\int \frac{dX}{(X^2 - 1)^{3/2}}$ Let $X = \sec \theta \rightarrow dX = \sec \theta \tan \theta d\theta$, $\sqrt{X^2 - 1} = |\tan \theta| = \tan \theta$
 $\theta \in [0, \frac{\pi}{2})$ (from $X > 1$) $\theta = \operatorname{arcsec} X$

$$\rightarrow \int \frac{\sec \theta \tan \theta d\theta}{\tan^3 \theta} = \int \frac{\cos \theta}{\sin^2 \theta} d\theta = \int \frac{1}{u^2} du \quad (u = \sin \theta) = -\frac{1}{u} + C = -\frac{X}{\sqrt{X^2 - 1}} + C$$



$$\int \frac{1}{X^3 + 1} dX : \frac{1}{X^3 + 1} = \frac{A}{X + 1} + \frac{Bx + C}{X^2 + X + 1} = \frac{1}{3} \left(\frac{1}{X + 1} + \frac{-X + 2}{X^2 + X + 1} \right)$$

$$1 = A(X^2 + X + 1) + (Bx + C)(X + 1) \quad (X - \frac{1}{2})^2 + \frac{3}{4}$$

$$A + B = 0, -A + B + C = 0, A + C = 1$$

$$2B + C = 0 \quad C = \frac{2}{3}$$

$$-B + C = 1 \quad B = -\frac{1}{3}, A = \frac{1}{3}$$

$$u = (X - \frac{1}{2}) \cdot \frac{2}{\sqrt{3}}$$

$$\rightarrow X^2 + X + 1 = \frac{3}{4}u^2 + \frac{3}{4}$$

$$X = \frac{1}{2} + \sqrt{\frac{3}{4}u} \quad dX = \sqrt{\frac{3}{4}} du$$

$$\int \frac{-X + 2}{X^2 + X + 1} dX = \int \frac{\frac{2}{\sqrt{3}} \left(\frac{Au + B}{u^2 + 1} \right) \cdot \frac{\sqrt{3}}{2} du}{\frac{3}{4}u^2 + \frac{3}{4}}$$

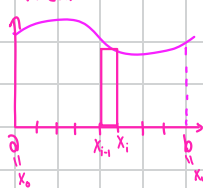
Substitution

8.7 Numerical Integrations

Motivation: $\int e^{-x^2} dx = ?$ (Agony!: cannot find antiderivative of e^{-x^2})

Resolution: Use numerical approximation

Recall



$$\int_a^b f(x) dx = \lim_{\|P\| \rightarrow 0} \sum_{i=1}^n f(c_i) \Delta x_i \quad \text{if the limit exists and is independent of choices of } c_i \text{ and } p's$$

$$\Delta x_i = x_i - x_{i-1}, \|P\| = \max(\Delta x_1, \Delta x_2, \dots, \Delta x_n)$$

$$\int_a^b f(x) dx \approx \sum f(c_i) \Delta x_i \quad \text{if } \|P\| \approx 0$$

Special cases: Divide $[a, b]$ into n subintervals of equal length: $\Delta x_i = \frac{b-a}{n}$

1. Left sum: $c_i = x_{i-1}; i=1, 2, \dots, n \rightarrow L_n = \sum_{i=1}^n f(x_{i-1}) \Delta x$

2. Right sum: $c_i = x_i; i=1, 2, \dots, n \rightarrow R_n = \sum_{i=1}^n f(x_i) \Delta x$

3. Middle sum: $c_i = \frac{x_i + x_{i-1}}{2}; i=1, 2, \dots, n \rightarrow M_n = \sum_{i=1}^n f\left(\frac{x_i + x_{i-1}}{2}\right) \Delta x$

Q: How good are approximations?

Error: $E_n^L = \int_a^b f(x) dx - L_n$
 $E_n^R = \int_a^b f(x) dx - R_n$
 $E_n^M = \int_a^b f(x) dx - M_n$

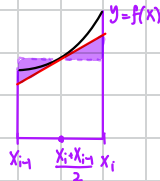
Theorem Suppose f has derivative on $[a, b]$ with f' contin on $[a, b]$, then

(i) $\left. \begin{matrix} E_n^R \\ E_n^L \end{matrix} \right\} \leq \frac{(b-a)^2}{2n} \max_{[a,b]} |f'(x)| \rightarrow 0$
 $= O\left(\frac{1}{n}\right)$ as $n \rightarrow \infty$

(ii) $E_n^M \leq \frac{(b-a)^3}{24n^2} \max_{[a,b]} |f''(x)| = O\left(\frac{1}{n^2}\right) \rightarrow 0$ as $n \rightarrow \infty$; f'' is contin on $[a, b]$

→ Midpoint rule better than left/right sums

Remarks: 1. Using midpoint rule is the same as using T-line to approximate (Area of $\square$ = Area of $\triangle$)



Suppose f is concave up on $[a, b]$
 $\therefore$ Area of $\triangle >$ Area of $\square$

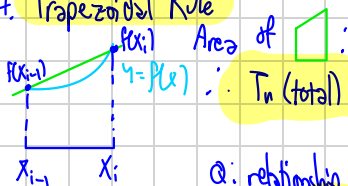
$\therefore \int_a^b f(x) dx > M_n$ → Underestimation of definite integral

Suppose f is concave down on $[a, b]$
 Area of $\square >$ Area of $\triangle$

$\therefore \int_a^b f(x) dx < M_n$ → Overestimation

More special Riemann's sum

4. Trapezoidal Rule



Area of $\square$: $T_n(\text{total}) = \sum_{i=1}^n \frac{f(x_{i-1}) + f(x_i)}{2} \Delta x$ → Easy to memorize

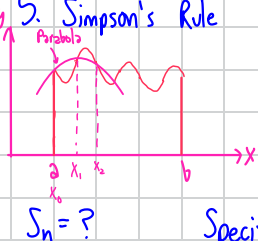
Q: relationship among L_n, R_n, T_n ?
 A: $T_n = \frac{L_n + R_n}{2} = \frac{1}{2} \sum_{i=1}^n f(x_i) \Delta x + \frac{1}{2} \sum_{i=1}^n f(x_{i-1}) \Delta x \rightarrow T_n$ better than L_n, R_n

In fact, $|E_n^T| \leq \frac{(b-a)^3}{12n^2} \max_{[a,b]} |f''(x)| = O\left(\frac{1}{n^2}\right) \rightarrow 0$ as $n \rightarrow \infty$

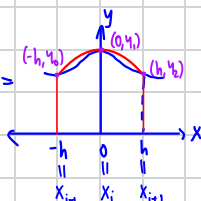
(M_n better, denominator of $|E_n^M| = 24n^2$)

5. Simpson's Rule

Basic fact: passing 3 pts on the plane $\exists$ one and only parabola



Special case =



want: area of $\square$

Let eq. of parabola be $y = Ax^2 + Bx + C$

$$\therefore y_0 = A(-h)^2 + B(-h) + C \quad \text{--- } \odot$$

$$y_1 = 0 + 0 + C \quad \text{--- } \star$$

$$y_2 = Ah^2 + Bh + C \quad \text{--- } \ominus$$

$$\text{Area of } \square = -h \int_{-h}^h (Ax^2 + Bx + C) dx = \left[\frac{Ax^3}{3} + \frac{Bx^2}{2} + Cx \right]_{-h}^h$$

$$= \frac{2Ah^3}{3} + 2Ch = \frac{h}{3}(2Ah^2 + 6C)$$

$$\therefore y_0 + y_2 = 2Ah^2 + 2C$$

$$= 2Ah^2 + 2y_1$$

$$S_n = \frac{\Delta x}{3} [(y_0 + 4y_1 + y_2) + (y_2 + 4y_3 + y_4) + \dots + (y_{n-2} + 4y_{n-1} + y_n)]$$

(n even)

$$= \frac{\Delta x}{3} [y_0 + 4y_1 + 2y_2 + 4y_3 + \dots + 2y_{n-2} + 4y_{n-1} + y_n]$$

$$|E_n^S| \leq \frac{(b-a)^5}{180n^4} \max_{[a,b]} (f^{(4)}(x)) = O\left(\frac{1}{n^4}\right) \rightarrow 0 \text{ as } n \rightarrow \infty$$

If n large,

Simpson	Midpoint	Trapezoid	Left	Right
Best			Worst	Worst

$$\int_0^1 x^2 dx = \left. \frac{x^3}{3} \right|_0^1 = \frac{1}{3}$$

$$L_{12} = 0.2928...$$

$$R_{12} = 0.3761...$$

$$T_{12} = 0.3344...$$

$$M_{12} = 0.3327...$$

$$S_{12} = 0.3333...$$

Q: Relationship of S_n to T_n & M_n

$$S_n = \frac{1}{3}(2T_n + M_{\frac{n}{2}})$$

8.8 Improper Integral

Recall: existence $\int_a^b f(x) dx$ needs: (i) $f(x)$ bounded on $[a, b]$ (not going to heaven/hell)

(ii) $[a, b]$ finite

Def $\int_a^b f(x) dx$ is called "Improper integral" if either (i) or (ii) is violated

Type I Improper integral: $f(x)$ is unbounded ex. $\int_0^1 \frac{1}{\sqrt{x}} dx$ ($\frac{1}{\sqrt{x}} \rightarrow \infty$ when $x \rightarrow 0$)

Type II Improper integral: a or b is $\infty, -\infty$ or $\int_a^b f(x) dx$ diverges ex. $\int_0^\infty e^{-x} dx$ (b is ∞)

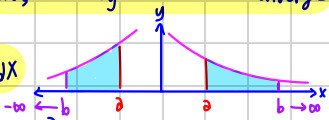
$$\int_0^\infty \sin x dx$$

(NOT WELL-DEFINED)

Worrying about convergence/divergence of improper integral

Def 1. For $\int_a^\infty f(x) dx$: If $\lim_{b \rightarrow \infty} \int_a^b f(x) dx$ exists as a finite, we say $\int_a^\infty f(x) dx$ converges

and define $\int_a^\infty f(x) dx$ as $\lim_{b \rightarrow \infty} \int_a^b f(x) dx$



2. For $\int_{-\infty}^a f(x) dx = \lim_{b \rightarrow -\infty} \int_b^a f(x) dx$ if it exists $\rightarrow \int_{-\infty}^a f(x) dx$ converges

3. Consider $\int_{-\infty}^\infty f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^\infty f(x) dx$; pick c as any real number

If both converge $\rightarrow \int_{-\infty}^\infty f(x) dx$ converges

If one of them diverges, $\int_{-\infty}^\infty f(x) dx$ diverges

Example: 1. $\int_0^\infty e^{-x} dx = \lim_{b \rightarrow \infty} \int_0^b e^{-x} dx = \lim_{b \rightarrow \infty} [-e^{-x}]_0^b = \lim_{b \rightarrow \infty} (-e^{-b} + 1) = 0 + 1 = 1$

2. $\int_{-\infty}^\infty \sin x dx = \int_{-\infty}^0 \sin x dx + \int_0^\infty \sin x dx$ $\therefore \int_0^\infty \sin x dx = \lim_{b \rightarrow \infty} \int_0^b \sin x dx = [\cos x]_0^b = \lim_{b \rightarrow \infty} [\cos b - \cos 0]$
DNE (Not well-defined) DNE

HW 8.4, 8.5, 8.7

